



What the attacker can buy
residential proxy addresses (Mi et al., 2019); shared addresses behind carrier-grade NAT (Richter et al., 2016); CAPTCHA solutions (Motoyama et al., 2010; Dionysiou and Athanasopoulos, 2020); as many identities as it likes (Douceur, 2002)

Controls at the service
per-entity limits, CAPTCHA, proxy and country lists (v1, Section 3); risk-based scoring of logins (Wiefling et al., 2019); supervised per-request detection on labeled logs (Huh et al., 2025)
This study: twelve source-side steps and two label-free destination policies keyed on the number block, a send counter and outcome tests (Section 4)

Controls at the vendor
prefix and country blocking from an account's history, a per-number pumping risk score (Twilio Inc., 2026a, 2026c); sends per prefix in five-minute windows with a 24-hour quarantine (Zalisky and Odarchenko, 2026)

Where the money is made
revenue-share number ranges advertised by providers and shared among them; detection from call records and test-number intelligence at an operator (Sahin et al., 2017; Sahin and Francillon, 2021)